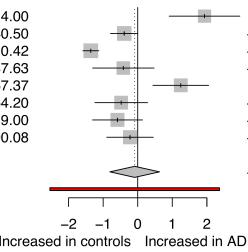
A − Dopamine

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Ionescu 1990 *	12	62.00	28.00	11	17.00	14.00	1.93	[0.91; 2.95]	11.1%
Alsen 2015 *	50	169.30	48.40	50	186.90	40.50	−0.39	[−0.79; 0.00]	13.7%
Xiong 2021 *	180	1.42	0.34	180	1.94	0.42	−1.36	[−1.59; −1.13]	14.0%
Nemet 2024 *	10	70.10	55.34	10	90.70	37.63	−0.42	[−1.31; 0.47]	11.7%
Rapoport 1978	15	211.72	0.24	14	151.87	67.37	1.24	[0.44; 2.05]	12.1%
Hanna 1996a	12	126.00	44.60	16	154.00	64.20	−0.48	[−1.24; 0.28]	12.3%
Hanna 1996b	15	19.70	7.10	16	24.60	9.00	−0.59	[−1.31; 0.14]	12.5%
Moriarty 2011	15	49.75	41.70	20	66.64	90.08	−0.22	[−0.90; 0.45]	12.7%
Random effects model							−0.09	[−0.81; 0.63]	100.0%
Prediction interval								[−2.54; 2.36]	

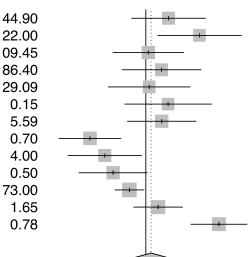
Heterogeneity: $I^2 = 91.6\%$, $\tau^2 = 0.9421$, $p < 0.0001$



B − Adrenaline

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Nagel−Hiemke 1984 *	21	90.50	89.70	9	50.70	44.90	0.49	[−0.30; 1.28]	7.4%
Ionescu 1990 *	12	122.00	89.00	11	43.00	22.00	1.15	[0.26; 2.05]	6.9%
Girardi 1995 *	17	229.00	111.32	11	223.00	109.45	0.05	[−0.71; 0.81]	7.6%
Oades 1998 *	14	111.90	86.40	9	81.70	86.40	0.34	[−0.51; 1.18]	7.2%
Nemet 2024 *	10	81.30	102.77	10	75.60	29.09	0.07	[−0.80; 0.95]	7.0%
Rapoport 1970	19	0.39	0.21	6	0.29	0.15	0.48	[−0.45; 1.41]	6.8%
Rapoport 1978	15	14.52	7.55	14	12.18	5.59	0.34	[−0.39; 1.07]	7.7%
Pliszka 1994	20	1.00	0.60	22	1.80	0.70	−1.20	[−1.86; −0.54]	8.0%
Hanna 1996a	12	4.10	1.50	16	7.00	4.00	−0.88	[−1.67; −0.09]	7.4%
Hanna 1996b	15	0.70	0.30	16	1.00	0.50	−0.70	[−1.43; 0.03]	7.7%
Anderson 2000	200	219.13	165.49	51	278.00	173.00	−0.35	[−0.66; −0.04]	9.4%
Konrad 2003	31	4.15	2.57	26	3.56	1.65	0.26	[−0.26; 0.79]	8.6%
Dvoráková 2007	57	10.13	5.74	17	2.09	0.78	1.57	[0.97; 2.17]	8.3%
Random effects model							0.11	[−0.32; 0.54]	100.0%
Prediction interval								[−1.48; 1.70]	

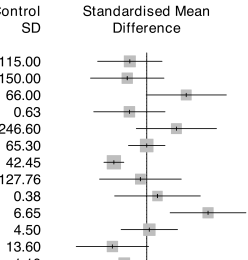
Heterogeneity: $I^2 = 81.1\%$, $\tau^2 = 0.4844$, $p < 0.0001$



C − Noradrenaline

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Mikkelsen 1981 *	15	219.00	116.00	18	262.00	115.00	−0.36	[−1.05; 0.33]	6.1%
Nagel−Hiemke 1984 *	21	241.00	186.00	9	317.00	150.00	−0.42	[−1.21; 0.37]	5.5%
Ionescu 1990 *	12	248.00	78.00	11	184.00	66.00	0.85	[−0.01; 1.71]	5.0%
Girardi 1995 *	17	1.55	0.62	11	1.79	0.63	−0.37	[−1.14; 0.39]	5.6%
Oades 1998 *	14	735.50	446.70	9	481.90	246.60	0.64	[−0.22; 1.50]	5.0%
Alsen 2015 *	50	232.00	67.30	50	232.00	65.30	0.00	[−0.39; 0.39]	8.0%
Xiong 2021 *	180	177.70	37.92	180	206.20	42.45	−0.71	[−0.92; −0.49]	9.0%
Nemet 2024 *	10	306.20	202.07	10	330.20	127.76	−0.14	[−1.01; 0.74]	5.0%
Rapoport 1970	19	1.13	0.38	6	1.04	0.38	0.23	[−0.69; 1.15]	4.7%
Rapoport 1978	15	31.21	12.84	14	17.19	6.65	1.32	[0.50; 2.13]	5.3%
Pliszka 1994	20	11.20	17.20	22	10.50	4.50	0.06	[−0.55; 0.66]	6.6%
Hanna 1996a	12	13.70	6.00	16	22.10	13.60	−0.74	[−1.52; 0.04]	5.5%
Hanna 1996b	15	2.30	0.90	16	2.80	1.10	−0.48	[−1.20; 0.23]	5.9%
Anderson 2000	200	931.56	480.44	51	1060.00	500.00	−0.26	[−0.57; 0.04]	8.5%
Konrad 2003	31	26.10	10.47	26	24.46	9.49	0.16	[−0.36; 0.68]	7.2%
Dvoráková 2007	57	22.51	9.97	17	15.55	4.37	0.76	[0.21; 1.32]	6.9%
Random effects model							−0.00	[−0.29; 0.28]	100.0%
Prediction interval								[−1.05; 1.04]	

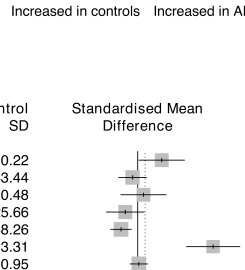
Heterogeneity: $I^2 = 76.3\%$, $\tau^2 = 0.2180$, $p < 0.0001$



D − Serotonin

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Irwin 1981 *	55	0.79	0.46	38	0.62	0.22	0.44	[0.02; 0.86]	12.2%
Comings 1990 *	127	92.81	55.39	109	97.61	43.44	−0.10	[−0.35; 0.16]	13.5%
Novkovic 2009 *	84	1.41	0.50	30	1.36	0.48	0.11	[−0.31; 0.52]	12.2%
Molina−Carballo 2013 *	136	20.18	20.40	42	25.22	25.66	−0.23	[−0.58; 0.12]	12.8%
Wang 2018 *	216	164.25	88.39	216	191.61	88.26	−0.31	[−0.50; −0.12]	14.0%
Jawad 2019 *	56	121.70	30.31	30	85.64	13.31	1.39	[0.90; 1.88]	11.5%
Raghavan 2022 *	269	0.04	1.01	326	0.02	0.95	0.02	[−0.14; 0.18]	14.1%
Moriarty 2011	15	152.59	59.47	20	157.15	46.06	−0.09	[−0.76; 0.58]	9.7%
Random effects model							0.13	[−0.23; 0.50]	100.0%
Prediction interval								[−1.09; 1.36]	

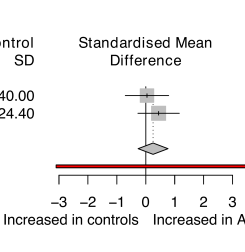
Heterogeneity: $I^2 = 85.3\%$, $\tau^2 = 0.2347$, $p < 0.0001$



E − 3,4−dihydroxyphenylacetic acid

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Hanna 1996a	12	405.30	215.80	16	397.60	140.00	0.04	[−0.71; 0.79]	47.7%
Hanna 1996b	15	93.90	45.60	16	77.40	24.40	0.44	[−0.27; 1.16]	52.3%
Random effects model							0.25	[−0.26; 0.77]	100.0%
Prediction interval								[−3.10; 3.60]	

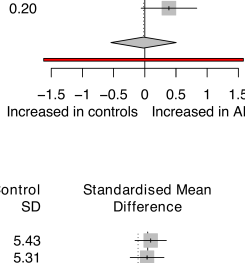
Heterogeneity: $I^2 = 0.0\%$, $\tau^2 = 0$, $p = 0.4472$



F − Homovanillic acid

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Rapoport 1978	15	4.17	1.89	14	3.73	1.39	0.26	[−0.48; 0.99]	17.8%
Shekim 1982	21	2732.00	710.00	12	3068.00	452.00	−0.52	[−1.24; 0.20]	17.9%
Shekim 1987	28	2579.00	783.00	23	3239.00	835.00	−0.81	[−1.38; −0.23]	20.5%
Kusaga 2002	37	7.44	4.95	21	5.21	3.25	0.50	[−0.05; 1.04]	21.0%
Bos 2015	40	1.50	0.30	39	1.40	0.20	0.39	[−0.06; 0.83]	22.7%
Random effects model							−0.02	[−0.54; 0.50]	100.0%
Prediction interval								[−1.62; 1.58]	

Heterogeneity: $I^2 = 75.0\%$, $\tau^2 = 0.2614$, $p = 0.0030$



G − 5−hydroxy−indoleacetic acid

Study	ADHD N	ADHD Mean	ADHD SD	Control N	Control Mean	Control SD	SMD	95% CI	Weight
Oades 2010a *	35	12.84	5.12	28	12.33	5.43	0.10	[−0.40; 0.59]	35.6%
Kusaga 2002	37	4.31	2.21	21	4.38	5.31	−0.02	[−0.55; 0.52]	34.6%
Moriarty 2011	15	1087.54	354.38	20	2093.16	1116.61	−1.12	[−1.84; −0.39]	29.8%
Random effects model							−0.31	[−1.02; 0.41]	100.0%
Prediction interval								[−3.18; 2.57]	

Heterogeneity: $I^2 = 74.9\%$, $\tau^2 = 0.3124$, $p = 0.0186$

